when the angle of inclination of the board is greater than the angle of repose. At the same time if inclination angle is much larger than the angle of repose, the slider will reach the bottom at greater speed and get hurt.

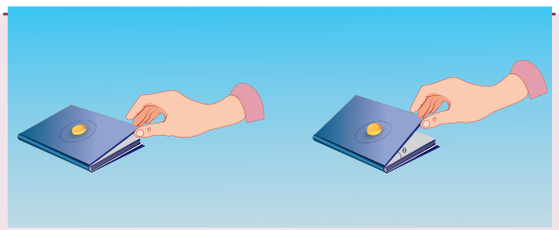


**Figure 3.31** Sliding board

## ACTIVITY

**Measuring the coefficients of friction**

Take a hard bound note book and a coin. Keep the coin on the note book. The note book cover has to be in an inclined position as shown in the figure. Slowly increase the angle of inclination of the cover with respect to rest of the pages. When the angle of inclination reaches the angle of repose, the parallel component of gravitational force ($mg \sin\theta$) to book surface becomes equal to the frictional force and the coin begins to slide down. Measure the angle of inclination and take the tangent of this angle. It gives the coefficient of static friction between the surface of the cover and coin. The same can be repeated with other objects such as an eraser in



order to observe that the coefficient of static friction differs from case to case.

**Note**

At the point of sliding $\tan\theta_s = \mu_s$

To measure the coefficient of kinetic friction, reduce the inclination of the book after it starts sliding, such that the coin/ eraser moves with uniform velocity. Now measure the angle from which coefficient of kinetic friction can be calculated as

$$\mu_K = \tan\theta_k$$

Observe that $\theta_K < \theta_S$

## 3.6.8 Rolling Friction

The invention of the wheel plays a crucial role in human civilization. One of the important applications is suitcases with rolling on coasters. Rolling wheels makes it easier than carrying luggage. When an object moves on a surface, essentially it is sliding on it. But wheels move on the surface through rolling motion. In rolling motion when a wheel moves on a surface, the point of contact with surface is always at rest. Since the point of contact is at rest, there is no relative motion between the wheel and surface. Hence the frictional force is very less. At the same time if an object moves



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